““Well, how bright does it need to be?”: Investigating the interplay of fluorescent signature and brightness in single-color unmixing controls

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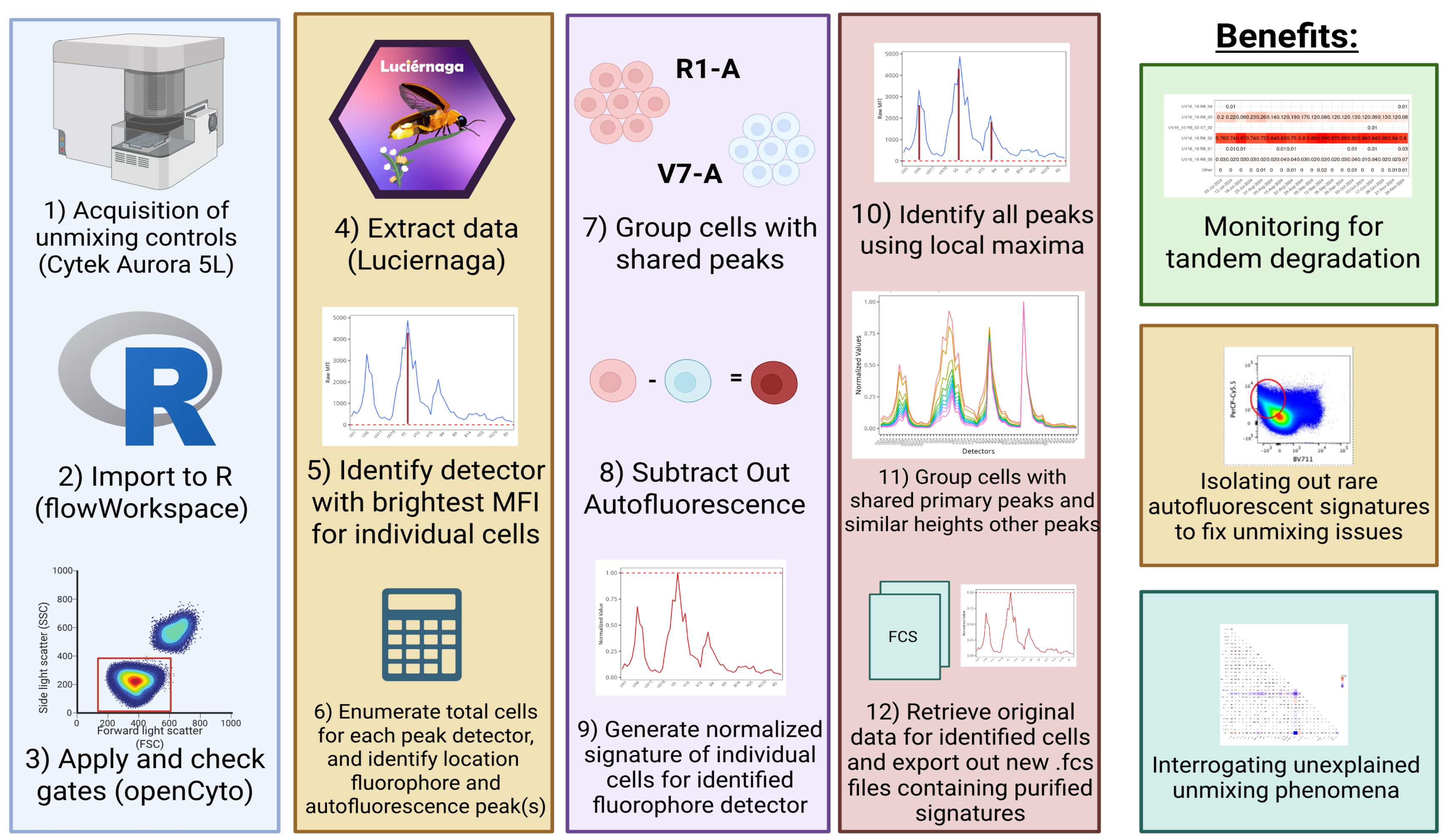
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Introduction

Spectral flow cytometry (SFC), with its capacity to resolve similar fluorophores and ability to rapidly acquire large number of events, enables more comprehensive phenotypic and functional analyses than conventional flow cytometry. Unmixing controls (both single-color and unstained) are critical for proper unmixing of high-dimensional panels, as their fluorescence signatures provision the reference matrix.
Discrepancies between the fluorescence signatures of unmixing controls and full-stained samples add uncertainty to the unmixing calculations and can lead to loss of resolution for individual fluorophores, (particularly in large panels with broader co-expression of markers) producing batch effects. These batch effects can affect performance, increase complexity, and impact reproducibility of unsupervised analysis methods that rely on median fluorescent intensity (MFI) for clustering, dimensionality visualization and normalization. While previous efforts at quality control have focused on factors related to instruments and/or full-stained samples, few tools exist to evaluate unmixing controls despite their critical importance. At the same time, the criteria for delineating a good unmixing control and re-using previous controls without affecting the unmixing remain loosely defined. We therefore set out to quantitatively assess how variation in the fluorescence signature and/or brightness of controls impacts unmixing of full-stained samples.

Methods


1) Acquisition of unmixing controls (Cytek Aurora 5L)
2) Import to R (flowWorkspace)
3) Apply and check gates (openCyto)
4) Extract data (Luciernaga)
5) Identify detector with brightest MFI for individual cells
6) Enumerate total cells for each peak detector, and identify location fluorophore and autofluorescence peak(s)
R1-A
V7-A
7) Group cells with shared peaks
8) Subtract Out Autofluorescence
9) Generate normalized signature of individual cells for identified fluorophore detector
10) Identify all peaks using local maxima
11) Group cells with shared primary peaks and similar heights other peaks
FCS
12) Retrieve original data for identified cells and export out new .fcs files containing purified signatures
Benefits:
Monitoring for tandem degradation
Isolating out rare autofluorescent signatures to fix unmixing issues
Interrogating unexplained unmixing phenomena

Results

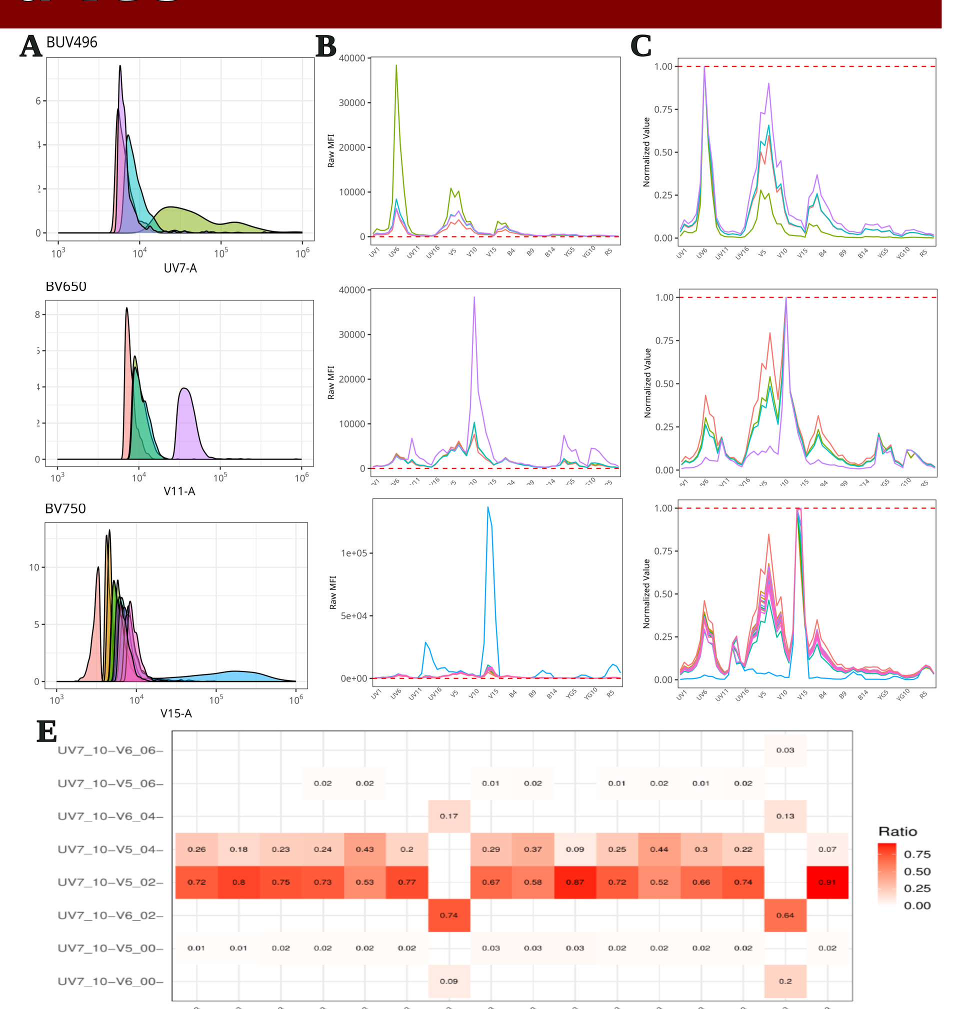


Figure 1. Greater number of signatures are present in dimmer staining cells compared to bright ones. (A)The density plots display the distribution of MFI values for cells that share a normalized signature, for BUV496 CD8 (top), BV650 CCR7 (middle) and BV750 IFNg (bottom). **(B)** The lineplot shows the median MFI for cells that share that normalized signature, for the given fluorophores. **(C)** The line plot depicts the median normalized signature for cells grouped on that basis **D)** The heatmap shows stability in fluorescent signature expression across a 15 experiment run for the BUV496 CD8 antibody. Individual dates are on the x-axis, with individual signatures on the y-axis. The of cells for a given signature to that of cells across all signatures is shown and shaded in red by value.

Results

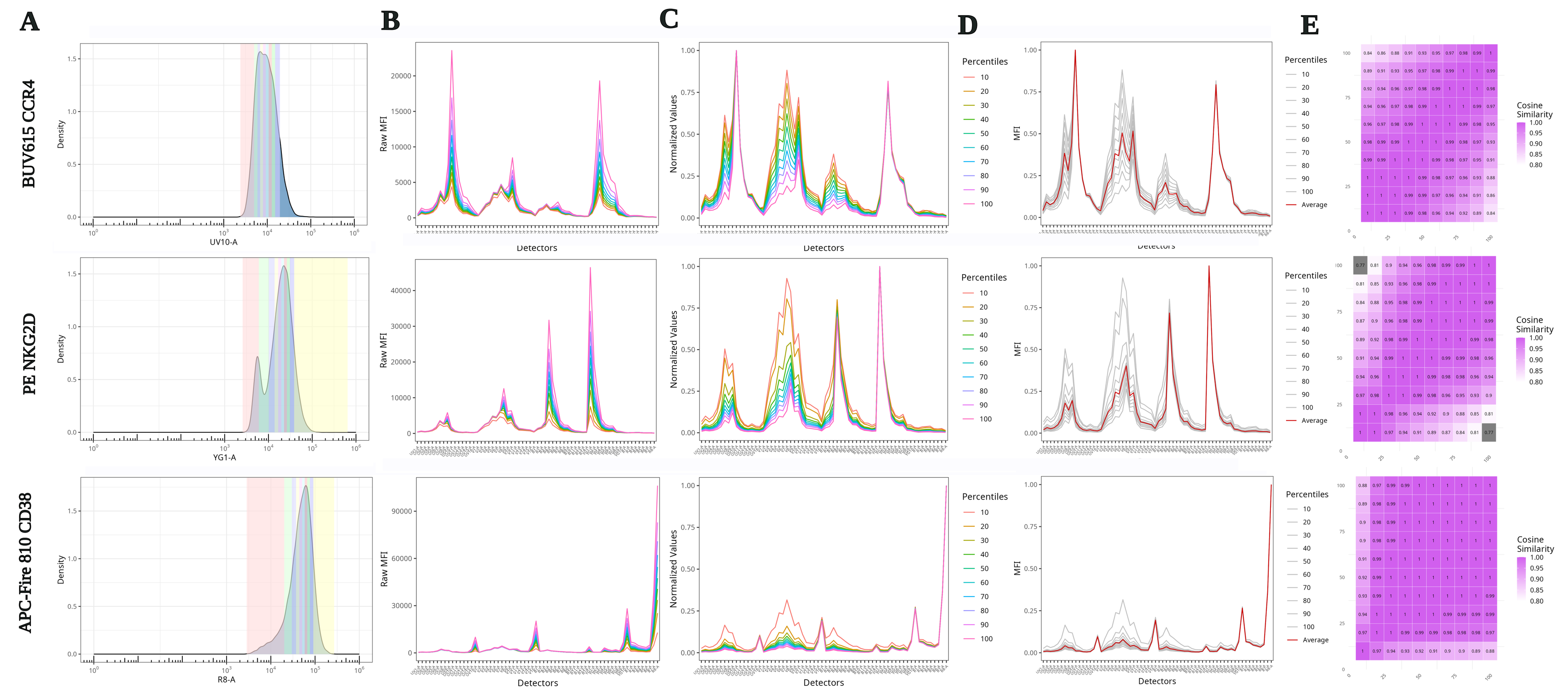


Figure 2. Residual autofluorescence after background subtraction contributes to a greater extent to the normalized signature of dimmer staining cells. A) The density plot displays bins separatating positive staining cells into 10 percentile bins, for BUV615 CCR4 (top), PE NKG2D (middle) and APC-Fire 810 CD38 (bottom) respectively. **B)** The line plots depict median fluorescent intensity (MFI) of cells within each bin depicted in A. **C)** The lineplots show the normalized MFI signatures of cells within each bin depicted in A. **D)** The lineplots display the average normalized signature of positive cells across all bins in red, and those of individual bins in grey. **E)** Cosine plots indicate how similar normalized signatures shown in C are.

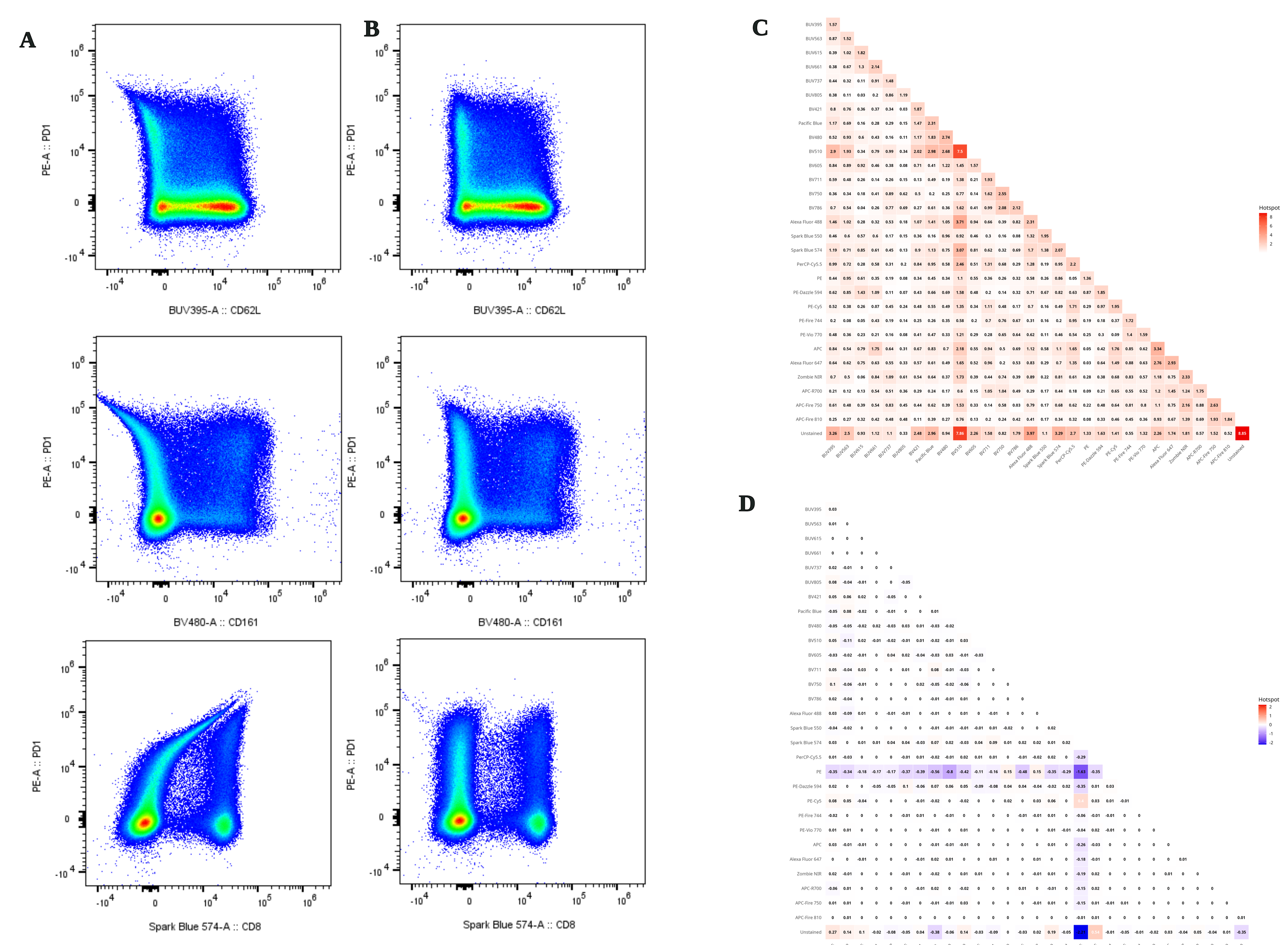


Figure 3. Unmixing with signatures of cells with lower MFI exacerbates unmixing-dependent spreading for given fluorophore and unstained control. The plots display iterative unmixing results for the full-stained sample, for the dimmer 10th percentile **(A)** and brighter 90th percentile **(B)** PE NKG2D signatures respectively. **C)** The triangular matrix depicts the unmixing-dependent spreading hotspots for the full-stained sample using the standard unmixing approach **D)** The triangular matrix depicts the difference between the hotspot matrices of the bright and dim PE NKG2D signatures used in the iterative unmixing.

Results

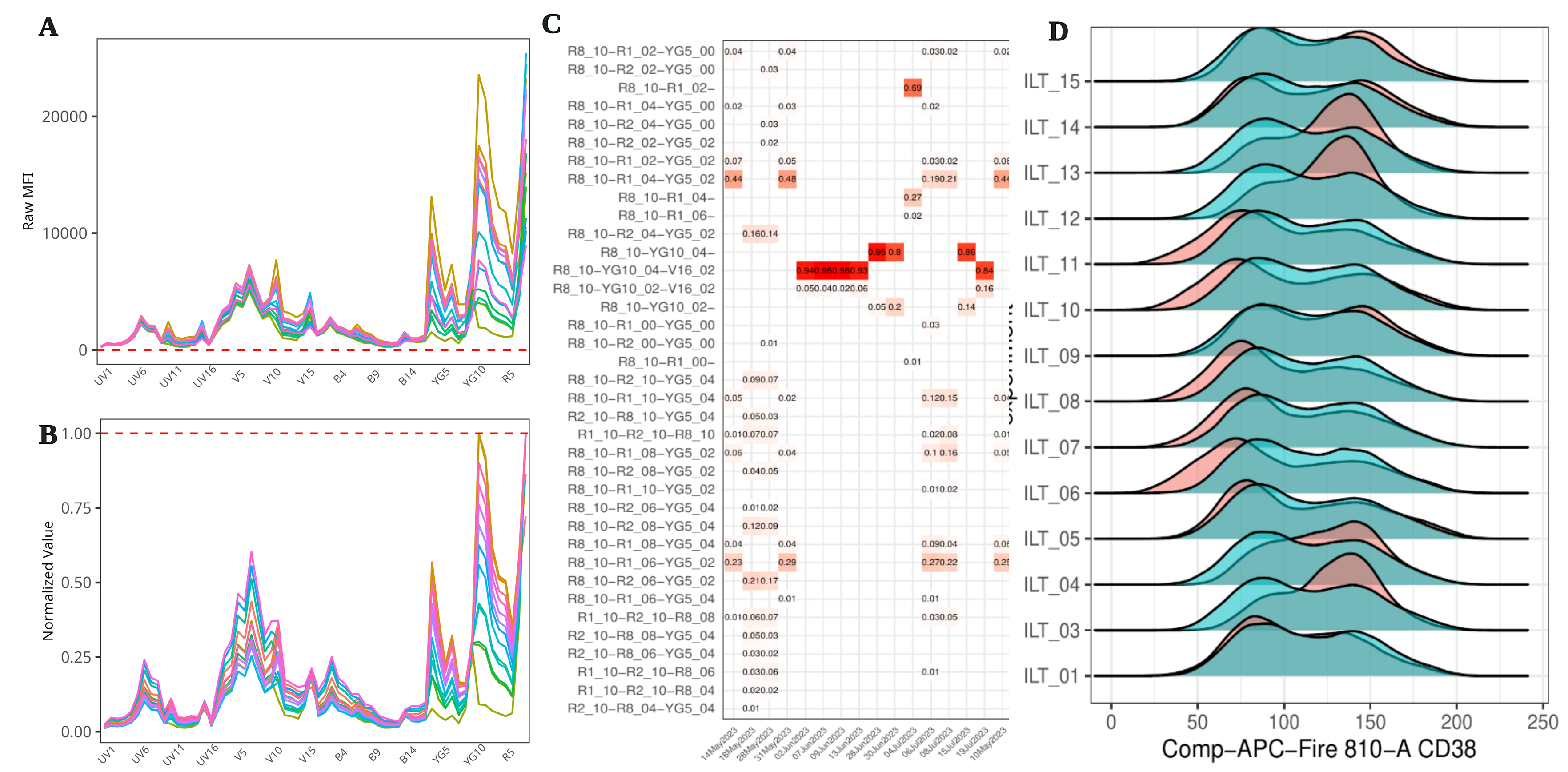
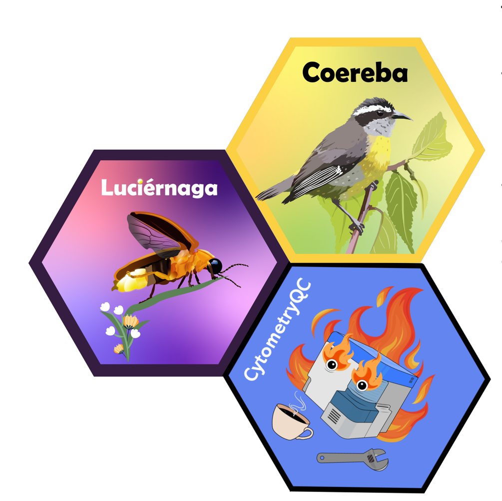


Figure 4. Aggregation of normalized signatures of individual cells fascilitates the detection of tandem degradation. A) The line plot shows the median MFI values for cells within signatures isolated from an APC-Fire 810 CD38 single color unmixing control. **B)** The lineplot depicts the median normalized signature for cells within the signatures isolated from the same unmixing control. **C)** The heatmap shows the stability of the APC-Fire 810 CD38 control across an experimental run, with individual dates on the x-axis, and isolated signatures on the y-axis, with ratio of cells within that signature compared to cells across all signatures shown in red. **D)** The ridgeplot shows the expression of APC-Fire 810 CD38 in the unmixed full-stain samples of the normalization control donor on the x-axis, with individual experiment days on the y-axis. Colors depict pre (teal) and post (red) normalization with CytoNorm V2.

Takeaways

Dimmer staining cells contain more fluorescence signatures after background subtraction compared to brighter staining cells. After background subtraction, residual autofluorescence signal is a greater contributor to the signatures of dimmer cells than brighter ones. This additional overlap with autofluorescence results in unmixing issues, typically tracking unmixing-dependent spreading hotspots. However, once a signature is isolated, gating on brighter staining cells doesn't change the unmixing outcome, just the scaling of the plots. Furthermore, profiling normalized fluorescent signatures of individual cells allows easier detection of tandem degradation, allowing for preventative measures to be taken early in an experimental series.

Data availability


Coereba
Luciérnaga
CytometryQC

Would you like to do this analysis with your own dataset? The R code used for the analysis and figure generation is available on our GitHub (https://github.com/DavidRach/SingleColors_Cyto2025). The single color and unstained unmixing controls will also be accessible via ImmPort (SDY3080) in the next release cycle.

The Luciernaga R package (and two other spectral flow cytometry focused projects) are in beta development and available via GitHub (https://github.com/DavidRach/Luciernaga). We are currently editing the documentation and plan to submit them to Bioconductor in the fall.

Take some stickers and track our progress!

Acknowledgements

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